

Risk matrix example

Severity →		1 Negligible	2 Minor	3 Moderate	4 Major	5 Catastrophic
↑ Likelihood	5 Very likely	5 Team member out sick	10 Complaint against company	15	20 Major competitor opens	25
	4 Probable	4 Temporary power outage	8	12	16 Data breach where client info is stolen	20
	3 Possible	3	6 Supply chain disruption	9 Team member fraud	12	15 Lawsuit filed against company
	2 Not likely	2	4	6	8 Significant loss of staff	10 Major flood or fire
	1 Very unlikely	1	2	3	4	5 Extended internet loss for entire company

Risk Register: Safeguarding Our E-Commerce Launch

A pragmatic look at potential threats to the successful delivery and operational stability of our new electrics e-commerce platform.

- Audience: Project Managers and Technical Leads. Focus: Delivery and Operational Risk.

Top Infrastructure and Environment Risks

Addressing foundational technical issues is critical for long-term stability and security.

Environment Parity Failure (ID 1)

Staging and Production environments are not identical, leading to undetected, production-only issues.

- Severity: High
- Owner: QA Lead, DevOps Engineer

Hardware Failure & Downtime (ID 2)

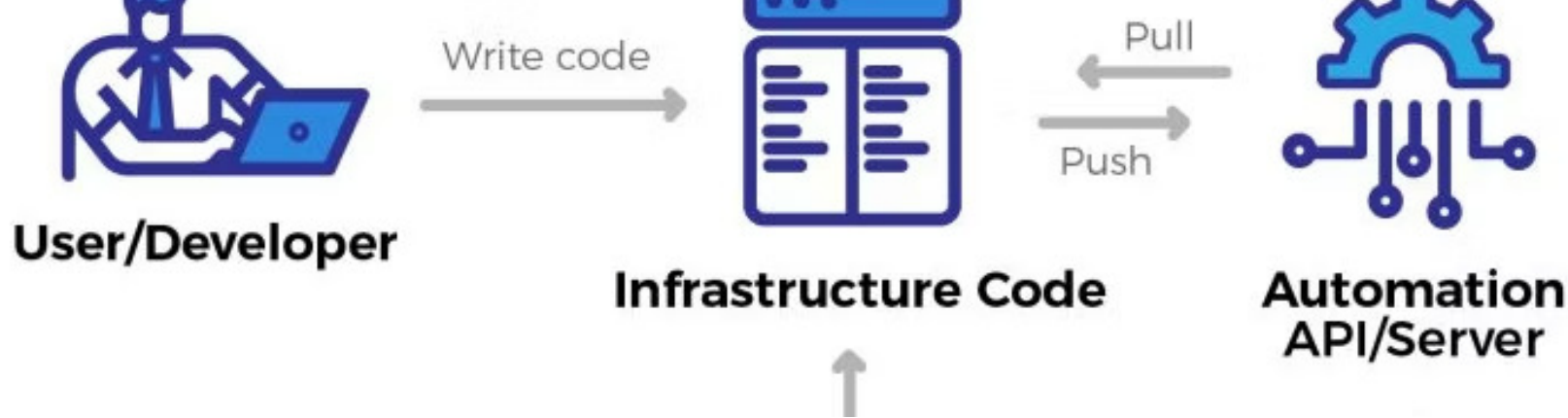
Hardware failure of self-hosted Kubernetes nodes causes service disruption and downtime.

- Severity: High
- Owner: DevOps Engineer

Internet Connection Disruption (ID 7)

Disruption affects self-hosted services, resulting in customer service interruption.

- Severity: High
- Owner: DevOps Engineer



Mitigation Strategy: Infrastructure & Environment



Implement IaC

Use Kubernetes manifests to ensure environment parity. Regular audits and documentation are key.



Backup & Monitoring

Regular backups, UPS power backup, and hardware health monitoring alerts. Keep spare components ready.



Redundancy & Failover

Implement automatic failover and document recovery procedures. Consider backup internet connectivity.

Security, Performance, and Data Integrity

Risks affecting user trust and system capacity as the product catalogue grows.

Authentication Security Flaws (ID 3)

Vulnerabilities in the authentication system could expose sensitive user data.

- Impact: High
- Owner: Software Developer

Database Degradation (ID 4)

Performance issues with 500+ products and high concurrent user traffic.

- Impact: High
- Owner: DevOps Engineer

Data Loss Risk (ID 13)

Inadequate backup procedures leading to irreversible data loss.

- Impact: High
- Owner: DevOps Engineer

Compliance Issues (ID 19)

Failure to comply with GDPR or other data protection regulations.

- Impact: High
- Owner: Project Manager



Mitigation: Ensuring Robust Systems



Security Protocols

Regular scans (SonarQube), use JWT/HTTPS, OWASP best practices, and penetration testing.



DB Performance

Implement indexing, query optimization, and connection pooling. Use Redis for crucial data caching.



Compliance & Data

Enforce data encryption (at rest/in transit). Implement consent banners and review Terms/Privacy policies.



Backup Strategy

Automated daily backups to external storage. Monthly recovery testing and verification logs.

Development Lifecycle and Quality Control

Addressing risks related to testing, technical debt, and integration complexity.

→ Insufficient Testing Coverage (ID 9)

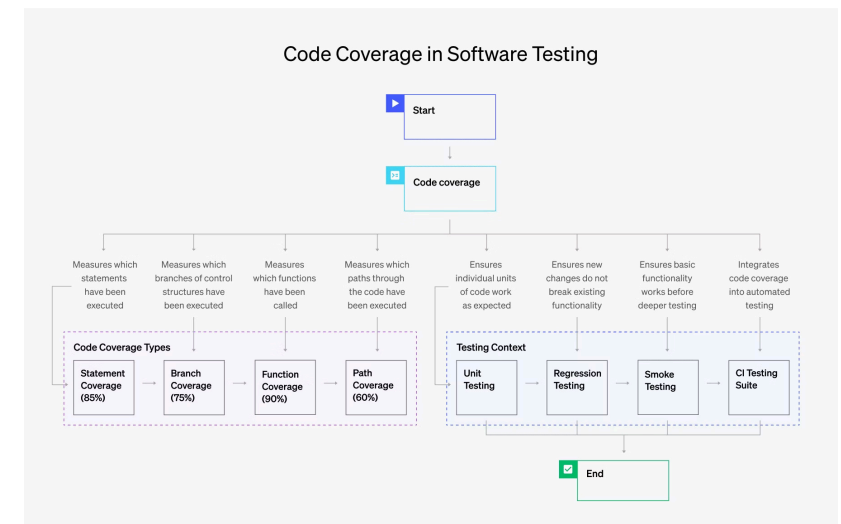
Less than adequate testing leads to critical bugs in the live production environment.
Mitigation: Enforce 80% code coverage (Codecov), automated testing (Playwright/K6), thorough UAT.

→ Technical Debt Accumulation (ID 10)

Rushed development creates code that is difficult to maintain and expand later.
Mitigation: Regular code reviews, dedicated refactoring time in sprints, SonarQube quality gates.

→ Payment Gateway Failures (ID 8)

VNPay integration issues or unannounced API changes disrupt sales transactions.
Mitigation: Comprehensive error handling, sandbox testing, API change subscription, payment abstraction layer.



Performance Optimization: Preventing Slow Load Times

Poor performance during peak load testing (>3 second page load) negatively impacts user experience (ID 12).

<3s

Target Page Load

Consistent monitoring is required to maintain sub-3 second page loads.

K6

Load Testing Tool

Regular performance testing using K6 throughout the development lifecycle.

CDN

Static Asset Delivery

Utilise Content Delivery Networks for image optimisation and static asset caching.

Mitigation includes: Lazy loading, image optimisation, code splitting, and regular performance analysis.

Project Management and Team Risks

Non-technical risks that threaten schedule, scope, and team capacity.

Key Personnel Unavailability (ID 5)

Illness or personal issues leading to critical resource bottlenecks.

Mitigation: Cross-train and establish backup responsibilities for key roles.

Knowledge Gaps / Expertise (ID 11, 16)

Lack of Kubernetes/microservices expertise or inadequate documentation.

Mitigation: Dedicated training, knowledge-sharing sessions, and mandated documentation (VitePress).

1

2

3

4

Scope Creep (ID 6)

Stakeholders requesting significant additional features mid-project.

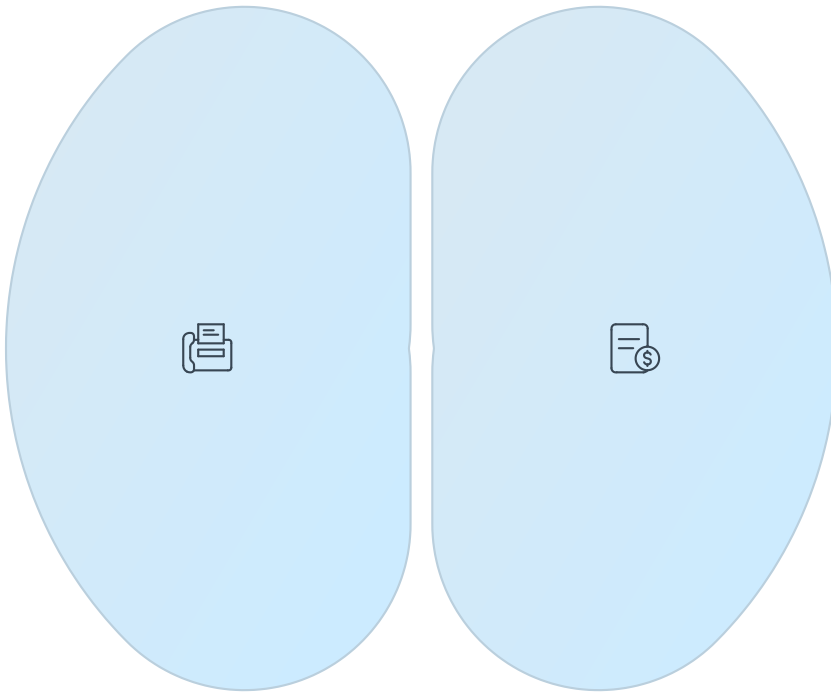
Mitigation: Enforce strict change control and manage expectations proactively.

Missed Project Deadline (ID 20)

Failure to meet the Week 12 launch commitment.

Mitigation: Agile methodology (2-week sprints), Jira tracking, critical path identification, scope adjustment if required.

Contingency Planning: Team and Budget Stability



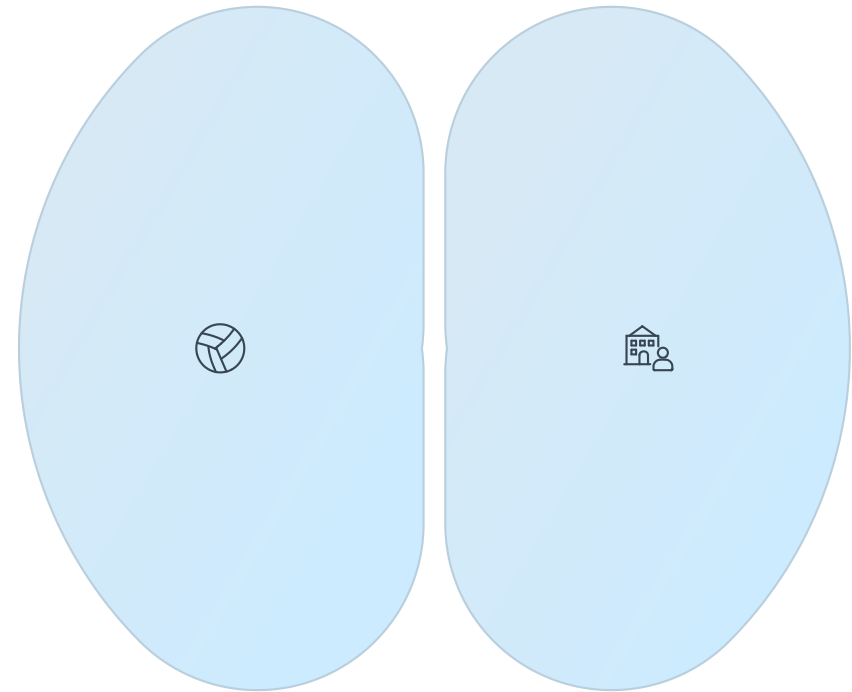
Communication

Daily stand-ups, real-time messaging, bi-weekly retrospectives.



Budget Overrun

Maintain \$750 contingency fund; weekly expense tracking; prioritize open-source.



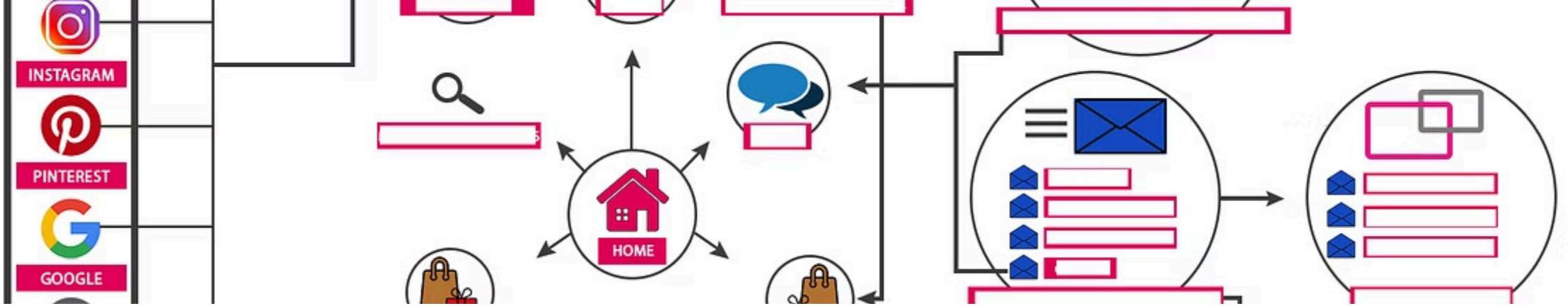
Skill Transfer

Formalised training and knowledge-sharing sessions to bridge expertise gaps.



Scope Control

Strict change control procedures to protect core delivery objectives.



Key Takeaways: Proactive Risk Management

Our success hinges on consistent application of mitigation strategies across all risk categories.

High Priority Action	Critical Performance Metric	Delivery Safeguard
Ensure Environment Parity (ID 1) via IaC to prevent production-only issues.	Prioritise DB Optimization (ID 4) and performance enhancements (ID 12) for scalability.	Strict Change Control (ID 6) and Testing Coverage (ID 9) protect the launch schedule (ID 20).

Next Steps: Project Manager to schedule [Mitigation Action Review](#) with Owners by end of week.